

TANG THANH TRUNG

Fresher Software Engineer

📍 Ho Chi Minh City | ✉ thanhtrung22052004@gmail.com | 📞 0773 735 100

🌐 Portfolio | 🔗 LinkedIn | 🐙 GitHub

SUMMARY

I am a **Software Engineer** with a strong focus on pursuing a career in Backend Development. I am a dedicated and fast learner, always eager to acquire new technologies and deepen my technical expertise. Having gained practical experience as a Backend Developer. I am now looking for a career in a professional environment where I can take on more difficult challenges and further refine my professional skills.

EDUCATION & CERTIFICATIONS

Saigon University	2022 – 2026
Engineer in Information Technology	GPA: 3.2/4.0
TOEIC 655 (Test of English for International Communication)	2025
Issued by IIG Vietnam (Official Representative of ETS)	

SKILLS

Languages: Java, JavaScript, Python, C#

Technologies: Spring Boot, FastAPI, .NET core, NestJS, ExpressJS, ReactJS, NextJS.

Databases & Storage: MySQL, PostgreSQL, MongoDB, Qdrant, ORM, Azure (Blob Storage)

Distributed Systems & Tools: Docker, Git, Firebase, Kafka

WORK EXPERIENCE

Backend Developer, Hephaestus Technology 🌐 12/2025 – present

Project: Infrasight – AI-Driven BIM Automation & Knowledge System

Participated in the full software development life cycle under the guidance of a Technical Mentor, Contributed to architectural discussions and directly implemented the following core features.

- **Domain-Specific AI Agent:** Developed a specialized AI Agent system for the Construction Industry to automate 3D BIM model modifications and generate technical construction reports using Python and FastAPI as the core backend orchestration layer.
- **Security:** Implemented **OAuth2 via Firebase** to manage secure identity and access control for enterprise project data.
- **Multi-Agent Orchestration:** Engineered a LangGraph-powered Supervisor-Worker architecture utilizing stateful loops and autonomous feedback nodes to orchestrate complex BIM tasks and resolve ambiguous requests through iterative reasoning.
- **Cloud-Native Service Integration:** Engineered an asynchronous pipeline to offload heavy 3D model processing to **Autodesk Platform Services (APS)**; developed a system to package and deploy C# AppBundles for automated execution on a headless Revit engine via the Design Automation API.
- **Asynchronous Architecture:** Integrated **SSE** for real-time reasoning streaming and **Background Tasks** to offload intensive BIM operations, ensuring a responsive, non-blocking system.
- **Knowledge-Driven RAG:** Integrated Qdrant Vector DB to provide agents with a high-fidelity retrieval layer, enabling grounded reasoning based on extensive BIM technical standards.
- **Tech Stack:** Python, FastAPI, LangGraph, Qdrant, Azure Blob Storage, Revit SDK (C#), PostgreSQL, OpenAI API.

PERSONAL PROJECTS

Job Recruitment Platform - Java & Spring Boot (Personal Project) 🌐 09/2025 – 12/2025

- **Platform Core:** Engineered a high-performance recruitment platform based on a **Microservices architecture** using **Java 21 and Spring Boot 3**; ensured system scalability and resilience through decoupled service boundaries and modern backend patterns.
- **Security:** Implemented a security layer using **Spring Security** and **JWT** (supporting **AccessToken** and **RefreshToken**). Applied **RBAC** (Role-Based Access Control) for fine-grained authorization and integrated **OAuth2** for seamless **Google Login**.

- **Distributed Messaging:** Orchestrated asynchronous communication between services using **Kafka** as a message broker to handle real-time job alerts and event-driven updates.
- **Real-time Interaction:** Developed a high-concurrency **Chat & Notification** module utilizing WebSockets, enabling low-latency communication between recruiters and candidates.
- **Tech Stack:** Java 21, Spring Boot 3, Spring Security, NextJS, OAuth2, JWT, Kafka, MySQL, WebSocket, Docker.

Job Recruitment - NodeJS & NestJS (Team Project)

05/2025 – 08/2025

- **Platform Core:** Engineered a high-performance recruitment platform using **NodeJS** and **NestJS** to manage comprehensive data for accounts, candidate profiles, recruiter entities, and the end-to-end application process.
- **Search & Discovery:** Implemented optimized search and filtering mechanisms, enabling candidates to find suitable job opportunities and empowering recruiters to efficiently identify potential talent.
- **Security:** Implemented a robust security layer using **Spring Security** and **JWT** (supporting **AccessToken** and **RefreshToken**). Applied **RBAC** (Role-Based Access Control) for fine-grained authorization.
- **Tech Stack:** NodeJS, NestJS, ReactJS, JWT, MongoDB, Docker, TypeScript.